

PETER ALEXANDER

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Mathematics Educator | Inquiry-Based Curriculum Design | Quantitative Modeling

Mathematics educator and curriculum designer with experience spanning advanced secondary, collegiate, and interdisciplinary instruction. Passionate about creating mathematical learning experiences that cultivate conceptual depth, intellectual curiosity, and creative inquiry. Experienced in designing rigorous curricula connecting abstract reasoning with computation, cryptography, finance, data science, and real-world quantitative modeling. Currently developing enrichment experiences in number theory, blockchain, and neural network modeling while mentoring student research and exploratory computational projects. Committed to fostering collaborative learning environments in which ambitious and diverse learners can develop mathematical confidence, independence, and intellectual maturity.

CORE COMPETENCIES

🔗 Inquiry-Based Mathematics Instruction 👥 Mentorship of Advanced & Gifted Learners

✍ User Oriented Curriculum Design & Assessment Development

✎ Expertise Across All Secondary & Collegiate Mathematics ☑ Phenomenological Representation and Abstraction

📊 Data Science, Quantitative Modeling & Analytics 📦 Linear Algebra, Number Theory & Cryptography

🔗 Computational Thinking & Mathematical Programming ⚙ Machine Learning & Neural Network Applications

🏗 Interdisciplinary STEM & Finance Integration 💻 Educational Technology & Interactive Assessment Systems

PROFESSIONAL EXPERIENCE

BASIS Independent Fremont

Fremont, California

Subject Expert Teacher

2025–Present

- Designed and taught rigorous inquiry-based mathematics curriculum spanning precalculus, linear algebra, quantitative finance, AP Statistics, and interdisciplinary STEM applications for advanced secondary learners.
- Developed original assessments, mathematical investigations, and interactive digital learning systems emphasizing conceptual understanding, proof-oriented reasoning, and mathematical communication.
- Integrated data science, computational modeling, and real-world quantitative reasoning into classroom instruction through exploratory and project-based learning experiences.
- Mentored students in advanced independent studies involving neural networks, machine learning, cryptography, blockchain technologies, and quantitative analysis.
- Designed advanced enrichment experiences in number theory, cryptography, and introductory blockchain concepts for mathematically ambitious students.
- Integrated DECA-oriented finance, economics, and analytics applications into mathematics instruction to connect abstract mathematical ideas with contemporary quantitative systems.
- Collaborated with students and faculty to cultivate intellectually ambitious, inquiry-centered classroom environments supporting both conceptual rigor and creative exploration.

Illumin Education

STEM Subject Area Specialist & Quantitative Consultant

Santa Clara, California

2024–Present

- Mentored advanced secondary and collegiate students across mathematics, physics, computer science, and quantitative reasoning, including intensive preparation for SAT, ACT, GRE, GMAT, AMC, and AIME examinations and competitions.
- Designed individualized enrichment pathways spanning AP Calculus, AP Statistics, AP Physics, AP Computer Science, multivariable calculus, quantitative finance, and advanced problem-solving curricula for mathematically ambitious learners.
- Guided students in exploratory projects involving neural networks, machine learning, cryptography, blockchain technologies, quantitative finance, and computational modeling.
- Consulted with educational stakeholders and strategic decision makers on the development of data-science-informed analytical frameworks and quantitative decision-support systems.
- Developed computational and statistical tools for educational analytics, modeling, and large-scale admissions research initiatives integrating Python, R, machine learning methodologies, and data visualization.
- Created instructional systems and interactive assessment environments supporting inquiry-based learning, mathematical experimentation, and real-world quantitative application.

COLLEGIATE & ADVANCED MATHEMATICS INSTRUCTION

University of California, Berkeley / UC Berkeley Extension / Laney College / College of Alameda Berkeley & East Bay, California

Instructor & Lecturer in Mathematics and Quantitative Studies

- Taught collegiate mathematics and quantitative reasoning courses including algebra, geometry, statistics, calculus, linear algebra, multivariable calculus, differential equations, topology, numerical analysis, and applied quantitative methods across university, extension, and community college environments.
- Designed instruction emphasizing conceptual understanding, abstraction, proof-oriented reasoning, and mathematical communication across foundational and advanced mathematical topics.
- Guided students entering STEM, quantitative finance, architecture, engineering, and technical disciplines through rigorous yet accessible mathematical instruction.
- Developed interdisciplinary instructional approaches connecting mathematics with computation, economics, science, architecture, and applied quantitative reasoning.
- Mentored students with varied academic backgrounds while maintaining high expectations for intellectual engagement and mathematical rigor.

California College of the Arts

Interdisciplinary Mathematics & Science Curriculum Development

San Francisco & Oakland, California

2003–2007

- Developed interdisciplinary mathematics and science curriculum tailored to art, architecture, and design students, including Geometry and Art, Precalculus for Architects, CAD-integrated quantitative instruction, introductory physics, and courses exploring the relationships among mathematics, science, art, and philosophy.

- Designed conceptually rich coursework emphasizing spatial reasoning, mathematical visualization, creative inquiry, and real-world application across artistic and technical disciplines.
- Collaborated across disciplines to create accessible yet intellectually rigorous quantitative learning experiences for diverse learner populations.
- Developed interdisciplinary mathematics and science curricula that became a foundational component of successful WASC accreditation efforts related to quantitative and scientific instruction.

EDUCATION

University of California, Berkeley

Berkeley, California

C.Phil., Mathematics

M.A., Mathematics

B.A., Mathematics

Fellowships & Honors

NSF Fellowship — Mathematics

NASA Space Sciences Fellowship — Solar Physics

TECHNICAL & QUANTITATIVE TOOLS

Programming & Quantitative Tools: Python, R, Java, SQL & NoSQL Databases, HTML/CSS/JavaScript, Git/GitHub, LaTeX, Statistical Modeling, Machine Learning, Data Visualization